

Active Directory Domain Services - Home Lab

Wednesday, September 16, 2026 3:54 PM

Active Directory Domain Services - Home Lab

Performed by: Thabang Norman Lesotho

Date: 16 September 2026

Platform: Oracle VirtualBox

Guest OS: Windows Server 2025 Standard Evaluation (Desktop Experience)

Documentation Tool: Microsoft OneNote (used to document the implementation process, configurations, and screenshots)

Documentation maintained in Microsoft OneNote. Screenshots were captured and inserted manually, as clipboard image copying between the host and the VM was not available.

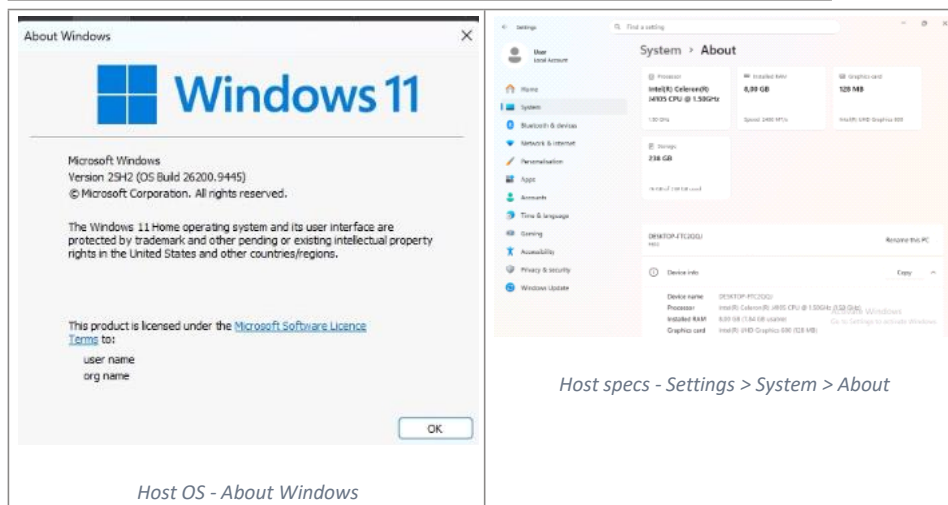
Objective

Build a functional Active Directory Domain Services (AD DS) lab environment using Windows Server 2025 and Oracle VirtualBox, for hands-on practice with AD DS, DNS, user and group management, Organizational Units (OUs), and Group Policy.

Environment Setup

Host Machine

Operating System	Windows 11 Home, Version 25H2 (Build 26200.9445)
Processor	Intel Celeron J4105 @ 1.50 GHz
RAM	8 GB
Storage	238 GB
Virtualization Platform	Oracle VirtualBox



Virtual Machine

VM Name	VMHomeLab
Operating System	Windows Server 2025 Standard Evaluation (Desktop Experience)
Memory Assigned	4096 MB
Processors	2
Virtual Disk	50 GB

EFI Enabled	Yes
Server Name	UDA-DC01
Domain / NetBIOS Name	ubuntudigital.local / UBUNTUDIGITAL

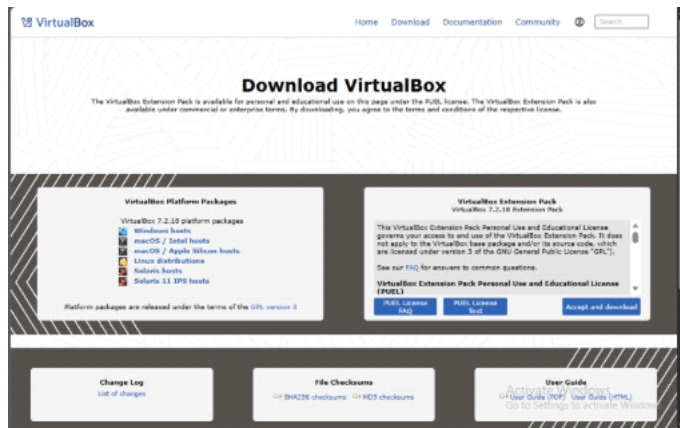
Step-by-Step Log

Step 1: Install Oracle VirtualBox

Action

- Downloaded Oracle VirtualBox.
- Installed using default settings and verified the installation.

Oracle VirtualBox installed successfully.

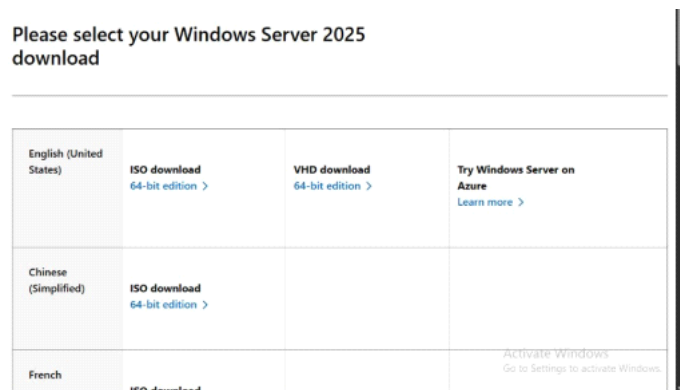


Step 2: Download Windows Server 2025 Evaluation ISO

Action

- Accessed the Microsoft Evaluation Center.
- Downloaded the Windows Server 2025 Evaluation ISO (64-bit).

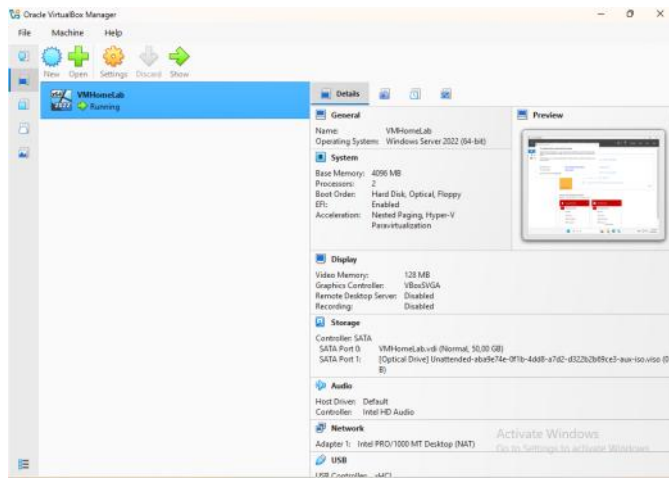
Windows Server installation media obtained successfully.



Step 3: Create the Virtual Machine

Action

- Created a new VM in VirtualBox using the specifications listed under Environment Setup above.

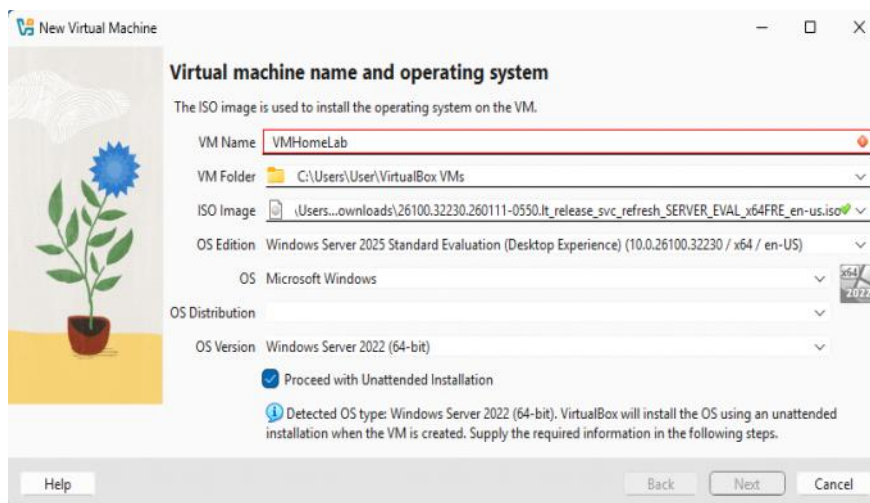


Virtual machine created successfully.

Step 4: Install Windows Server 2025

Action

- Mounted the ISO and ran setup.
- Selected Windows Server 2025 Standard Evaluation (Desktop Experience).
- Completed installation and configured the Administrator account.

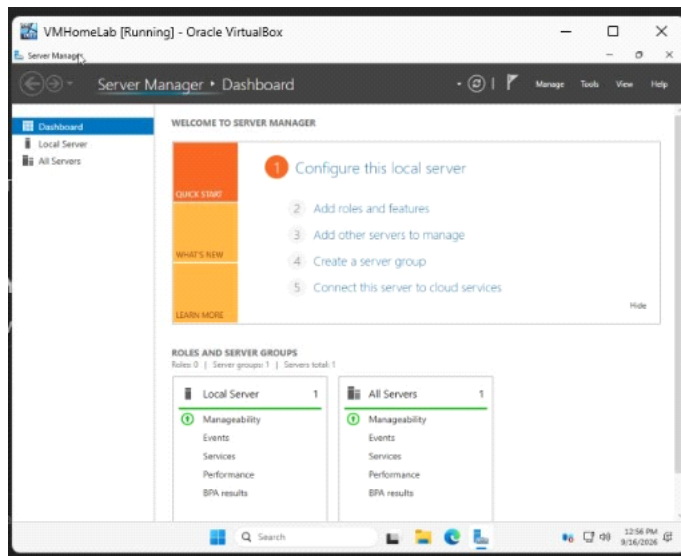


Windows Server installed successfully.

Step 5: Verify Server Manager

Action

- Logged into Windows Server and opened Server Manager.
- Reviewed the Dashboard, Local Server, and All Servers views.

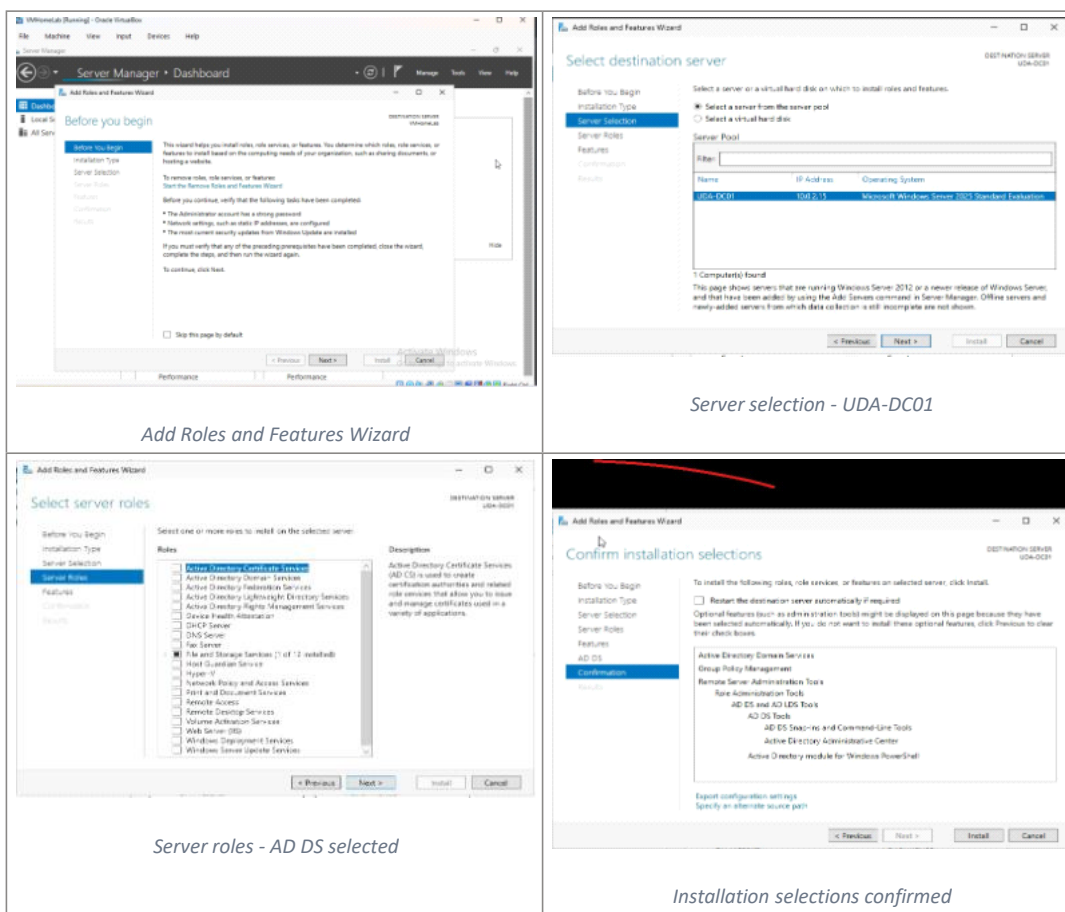


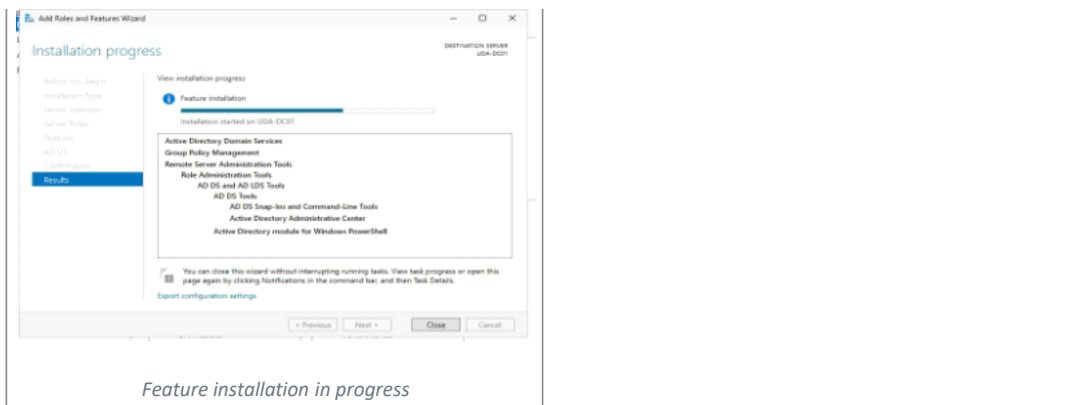
Server Manager accessible and functioning correctly.

Step 6: Install Active Directory Domain Services (AD DS)

Action

- Opened Server Manager > Manage > Add Roles and Features.
- Selected Role-based or feature-based installation, server UDA-DC01, and the Active Directory Domain Services role.
- Confirmed the automatically included tools and ran the installation.





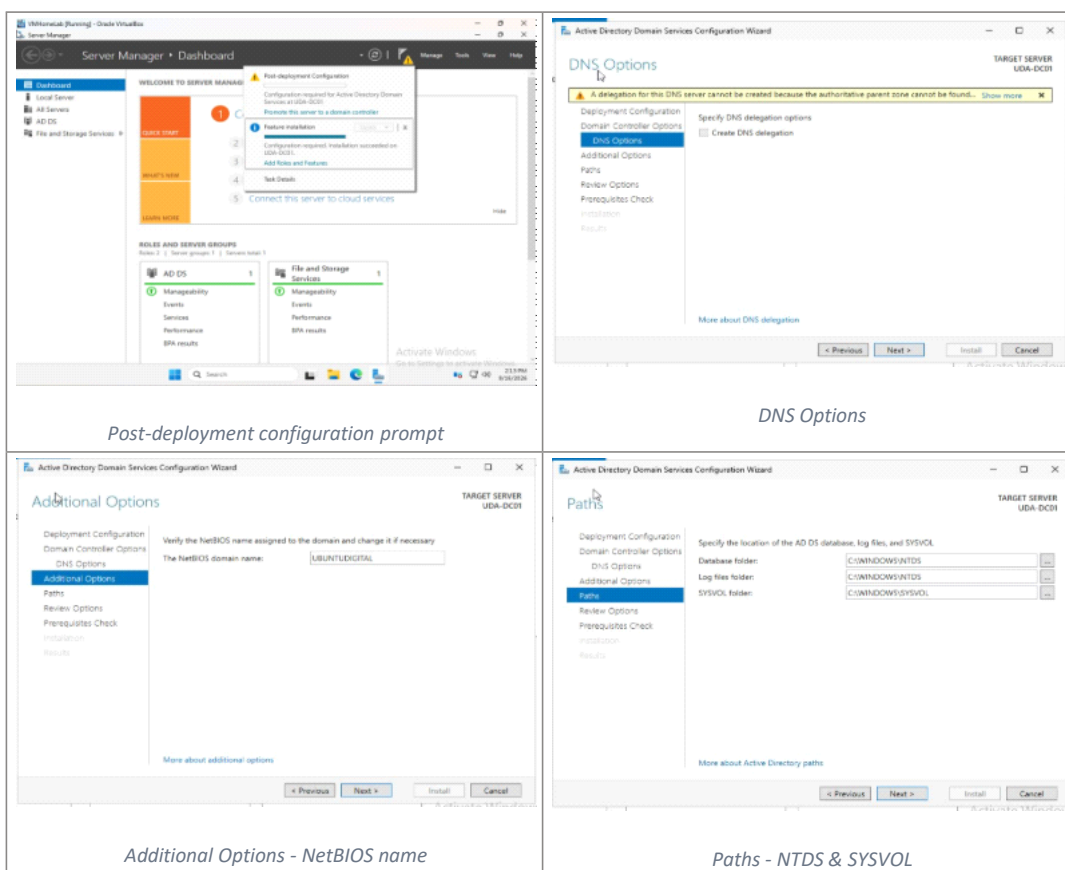
AD DS role installed successfully, along with Group Policy Management, RSAT, AD DS Tools, the AD Administrative Center, and the AD PowerShell module.

Step 7: Promote the Server to a Domain Controller

Action

- From the post-deployment notification, selected Promote this server to a domain controller.
- Deployment option: Add a new forest, with root domain ubuntu.digital.local.
- Configured DNS Server, Global Catalog, and NetBIOS name (UBUNTUDIGITAL).
- Accepted default paths for the database, logs, and SYSVOL (C:\Windows\NTDS / C:\Windows\SYSVOL).

Domain Controller configuration completed and ready for installation.

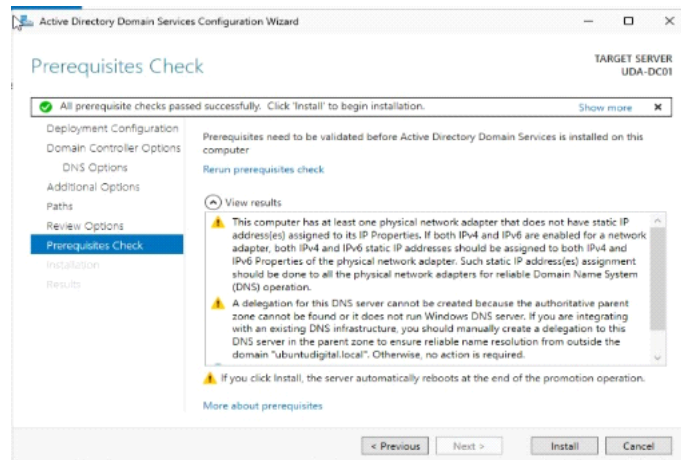


Step 8: Run Prerequisite Checks

Action

- Ran the built-in prerequisite validation before installation.
- No static IP address configured.
- DNS delegation could not be created.

All prerequisite checks passed. Two warnings were reviewed and accepted as expected for a lab environment:



Prerequisites check - passed

Step 9: Install the Domain Controller

Action

- Clicked Install and allowed the promotion process to run.
- The server installed AD DS and DNS, created the forest, configured SYSVOL, and rebooted automatically.

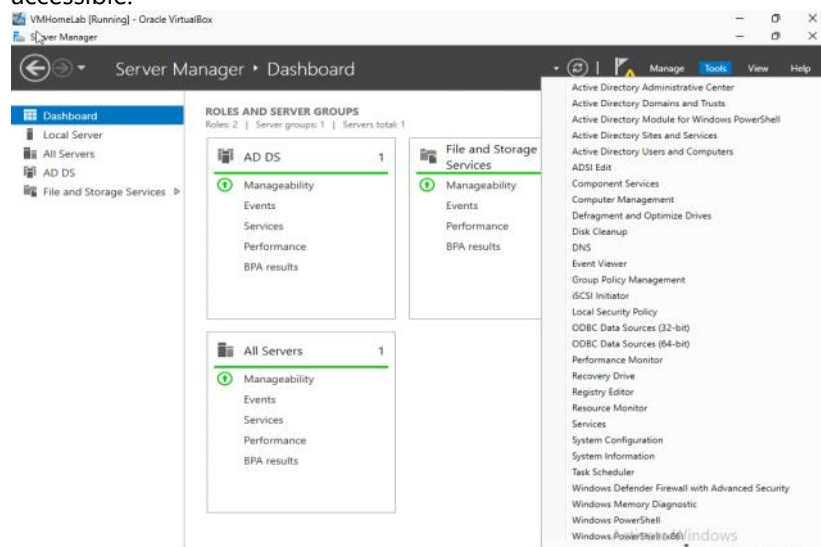
Domain Controller promotion completed successfully.



Step 10: Verify Active Directory Deployment

Action

- Opened Server Manager > Tools and confirmed all Active Directory components were installed and accessible.



Active Directory successfully deployed and Domain Controller confirmed operational.

Tools menu - all AD components present

Summary - Phase 1: Infrastructure Setup

Oracle VirtualBox installed

Windows Server 2025 installed

Virtual machine created (VMHomeLab)

Active Directory Domain Services installed

DNS installed

New forest created (ubuntudigital.local)

Domain Controller promoted

Administrative tools verified

Phase 1 status: Complete

Next Session - Phase 2: Active Directory Administration

Create Organizational Units

- Active Users
- Former Users
- Groups

Create Security Groups

- KZN_Learners
- IT_Support
- Former_Employees

Create Test Users

- Thabang Norman Lesotho
- Aaliyah Salaman
- Anele Sibiya
- Asanda Nkosi
- Chinonso Nwoye

Practice Tasks

- Create users
- Modify user properties
- Reset passwords
- Disable accounts
- Move users between OUs
- Manage group memberships